Sequential Logic – Part 4
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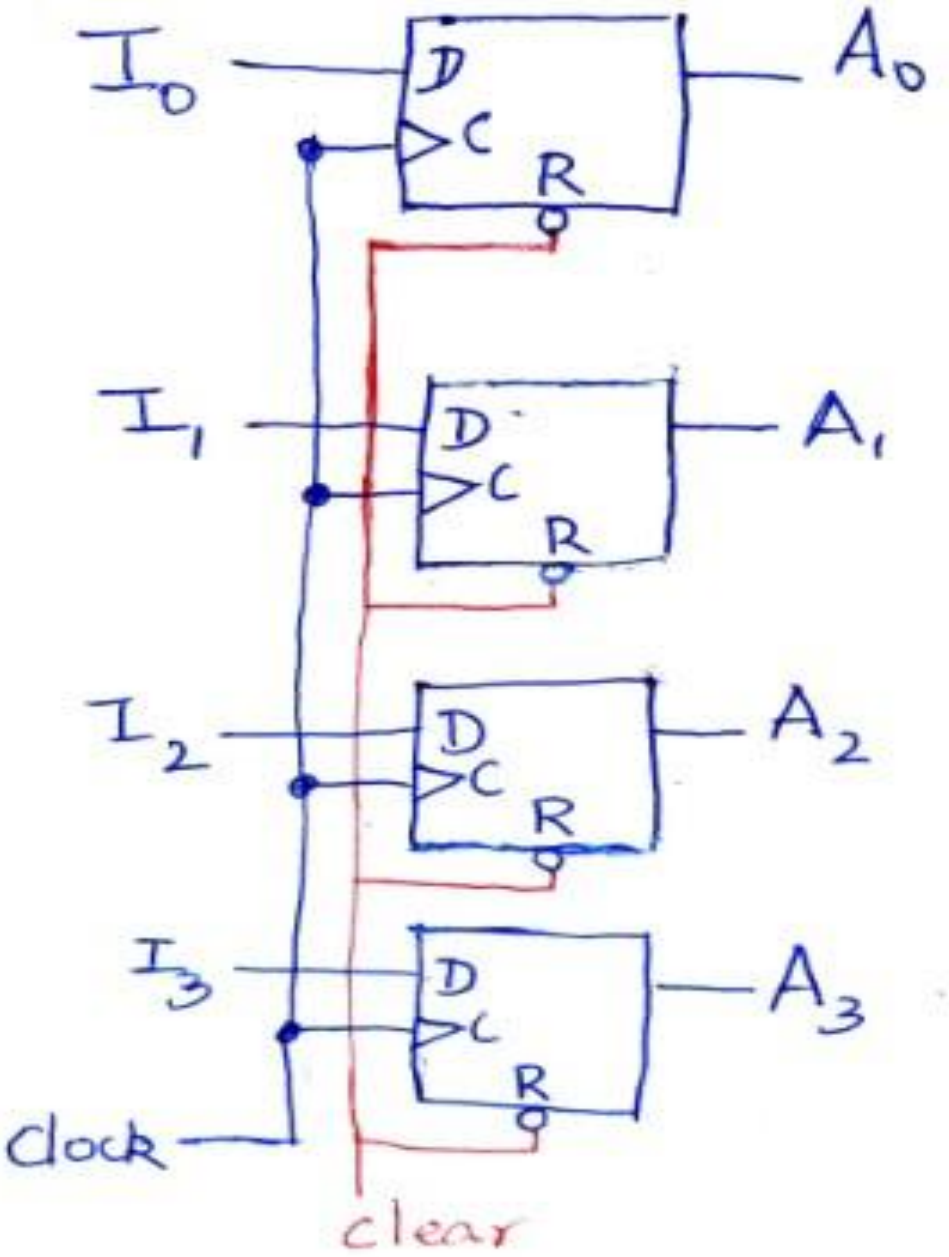
Review

- We have studied J-K flip-flops, setup time, hold time etc.
- We will now look at circuits that use many flip-flops to store information, perform counting etc.

Register

- One flip-flop stores one bit of information. Putting together several of them allows us to store multiple bits.
- For instance, when n flip-flops are used to n bits of information (say an n-bit number), we call these flip-flops as a register.
- A common clock is used for each flip-flop in a register. Many variations of the basic register notion are available. We will discuss these next.

Register with parallel load

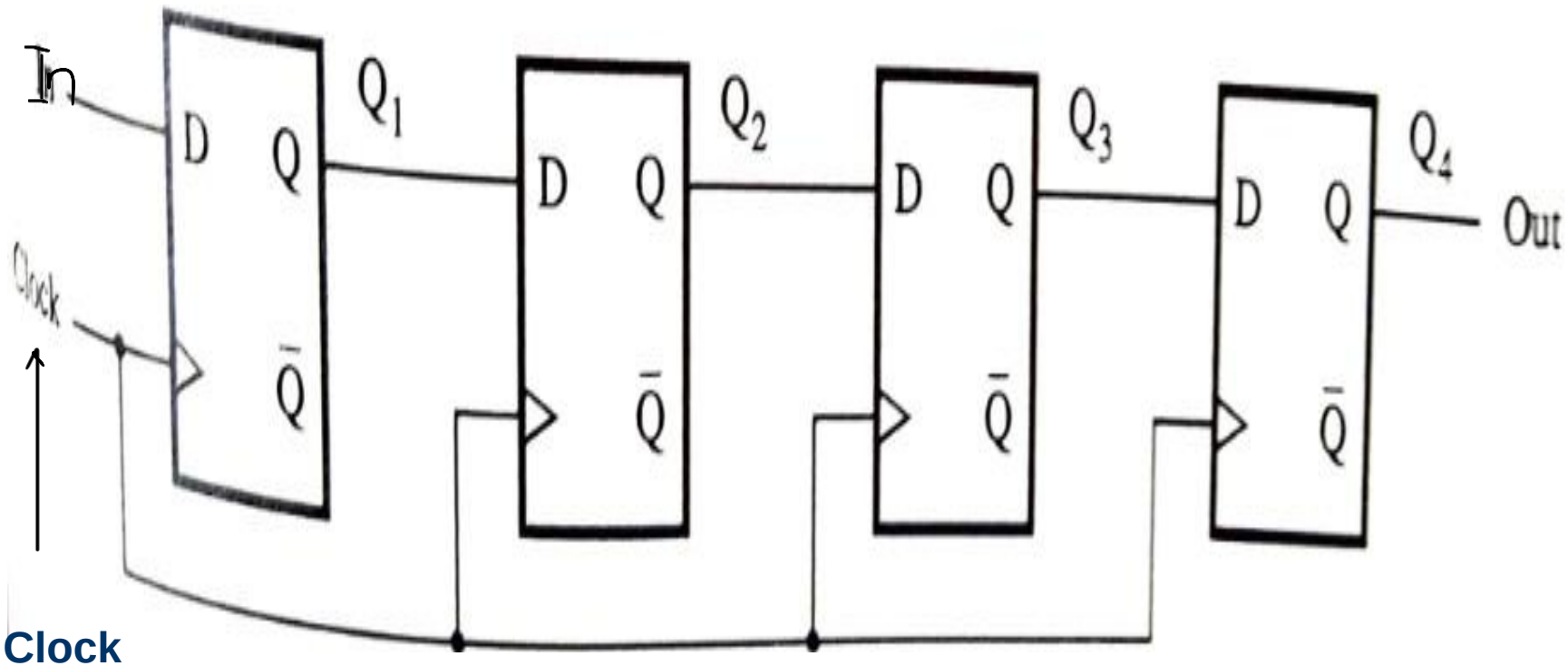


Explanation

- The binary data available at the four inputs (I_0, I_1, I_2 , I_3) get transferred into the register.
- The input Clear goes to the active low R (reset) input of all the four flip-flops
- A Clock input applied to the C inputs of the register will load all the four inputs in parallel.

Shift Register

D - flip flop



Clock

Explanation for the shift register

- The circuit shown takes inputs in serial fashion using the In input line.
- The contents of each flip-flop are transferred to the next flip-flop at each positive edge of the clock signal.
- Suppose the successive inputs are 1,0,1,1,1,0,0, and 0 and the initial state of the flip-flops is 0. Then we have the table shown next that captures what is available at the outputs of the various flip-flops during eight consecutive clock cycles.

Table showing the sequence

	In	Q_1	Q_2	Q_3	Q_4 = Out
t_0	1	0	0	0	0
t_1	0	1	0	0	0
t_2	1	0	1	0	0
t_3	1	1	0	1	0
t_4	1	1	1	0	1
t_5	0	1	1	1	0
t_6	0	0	1	1	1
t_7	0	0	0	1	1

Why do we need flip-flops ?

- We said we are using positive edge triggered flip-flops to realize a shift register. Why do we need these flip-flops ?

- QUESTION FOR STUDENTS

Why can’t we use a latch here ?

- That is, why cannot we have level-sensitive behaviour ?

Why do we need flip-flops ?

- To implement shift registers, we cannot use level-sensitive gated latches because a change in the value of In would propagate through more than one latch during the time the clock is equal to 1.
- We want the sequence shown in the table earlier.
- We need to therefore use edge-triggered flip-flops (or master-slave flip-flops).

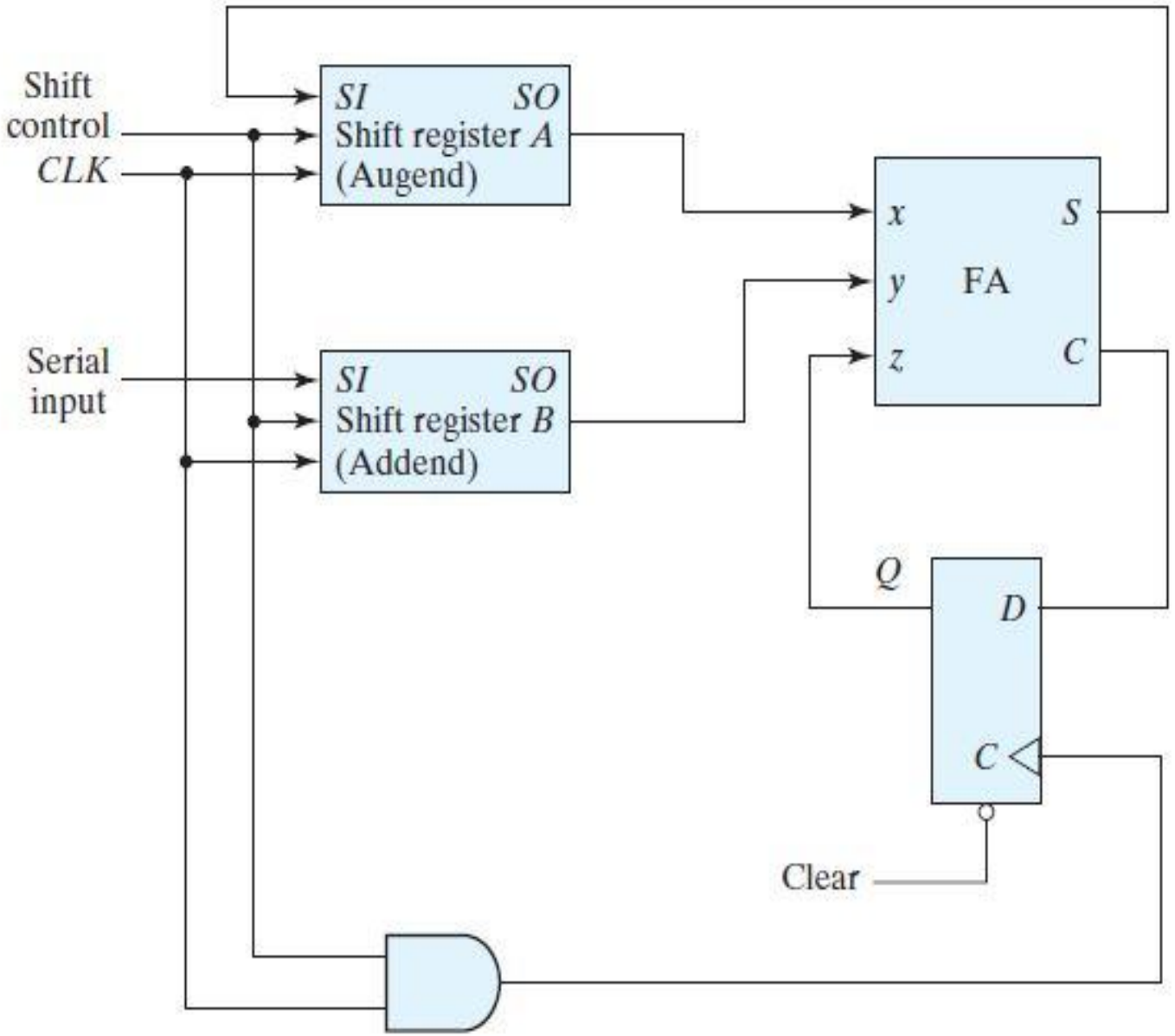
An application of shift register

- Shift registers have a variety of applications.
- One interesting application is serial addition.
- Recall we discussed parallel addition earlier (in carry lookahead adder for example).

More on serial addition

- In serial addition, we have a circuit that adds a pair of bits at a time through a single full-adder circuit. The pair comes from two shift registers that hold the augend and the addend.
- Augend: the number to which another is added
- Addend: the number to be added (second number)
- The circuit is shown next.

Serial adder



Overview of the serial adder

- Starting with the least significant pair of bits, the circuit adds one pair at a time through a single full adder (FA) circuit.
- The carry out of the full adder is transferred to a D flip-flop and the output of this flip-flop is used as the third (carry) input for the next pair of significant bits.
- Many questions need to be answered: What happens to the sum ? What is the role of the shift control signal ? How do we stop the addition ?

What happens to the sum (bit) ?

- The sum bit (from the S output of the full-adder) can, in principle, be transferred to a third register.
- This is, however, not efficient. Instead, we can shift the sum into A since anyway, bits in A (augend register) are getting shifted out during addition. So we can use just one register to store the augend as well as the sum bits.
- We assume the carry flip-flop is initialized to zero (observe the active-low Clear input to the D flip-flop).

What is the role of the Shift Control signal ?

- Earlier, we were managing the shifts via the clock signal. What exactly is the Shift Control input doing ?
- The Shift control signal enables both the registers and the carry flip-flop. Note the use of positive edge triggering for the D flip-flop and the shift control signal getting applied to this via an AND gate.
- When the clock pulse arrives, both the registers are shifted once to the right, the sum bit (S) from the full-adder enters the leftmost flip-flop of the shift register A while the carry output of the full-adder gets transferred to Q output of the flip-flop.

How do we stop the addition process ?

- Shift control enables the registers for 4 clock pulses if we have 4 bits in the registers.
- We stop the addition by disabling the shift control (input).
- Observe once again the output of the AND gate (one of whose inputs is the shift control signal) controlling the clock input to the D flip-flop that receives the carry bit from the full adder.

Extension to adding more numbers

- The arrangement shown can be used to perform A + B, A + B + C, A + B + C + D and so on.
- Note that the addend register bits are also getting shifted out during the first addition (of the initial pair in the augend and the addend registers).
- So the serial input of register B can be used to get in a new binary number.
- We can then perform the addition of contents in augend and addend registers again. This way, we can add three or more numbers and accumulate the result in register A.

What is the benefit of serial addition ?

- Question for students.

Benefit of serial addition

- The hardware requirement is reduced especially when we deal with addition of a large number of bits.
- This corresponds to lower silicon area on a chip.
- The drawback is, however, a reduction in speed.

Textbook and Reference

- M. Mano and M. Ciletti, Digital Design, Pearson Education, 2018
- S. Brown and Z. Vranesic, Fundamentals of digital logic with VHDL design, McGraw Hill, 2017.